

Spreading 48 Points in the Plane

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Abstract

We place 48 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 44.98168+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 49-point configuration gives 46.97026; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 48 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 44.98168\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 48$ is not on the page. Deleting a point from our 49-point configuration gives 46.97026; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n48.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
44981688224755090076123162
denominator:
999999999998142429725849
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

48 points, x and y per line. File: friedman-maxmin2d/coords/n48.txt.

```
11.878485813146 -10.488266593982      8.599352923201 -6.847455265083
11.882311752746 -8.102984144393      7.648252752126 -6.538572658611
13.740761615577 -7.991490090154      8.391303021655 -5.869337052153
8.484276826848 -9.575295181173      13.532711714031 -7.013371877223
10.484820827887 -6.219028702826      13.651143637034 -8.987466303686
10.276770926342 -5.240910489895      12.829585984221 -10.797149200454
11.573429146274 -9.054084315468      10.882419777875 -8.088285893223
7.135577705785 -7.397155384305      11.276662901213 -5.255608741066
13.045494785501 -6.140090900359      9.417084636413 -7.423054817904
11.060754099933 -9.912667041162      9.391626518603 -9.154918520462
12.227763072288 -5.564491347538      9.228193876989 -11.563180609982
9.971244146518 -10.893945003523      8.904409590072 -8.281637543597
12.391160859486 -9.629683868289      9.506702614956 -6.427078604372
9.904301564944 -8.296335794768      10.179294048064 -11.872063216454
12.302444515971 -6.809326506817      13.342261030561 -9.938566474761
11.395094824215 -7.229703167529      11.179186022935 -11.886761467625
9.153512433306 -10.318345450703      10.573537171402 -9.039386064298
12.789661444502 -7.682607483681      9.298652713411 -5.448960391441
10.060862125062 -9.897968789991      12.700043465959 -8.678583697214
8.086677876860 -7.706037990777      7.997059898317 -8.702014204309
7.045959727242 -8.393131597837      11.484712802759 -6.233726953997
7.741226557318 -10.244530787632      8.410462163777 -10.987581057162
10.971136121390 -10.908643254694      12.086535714691 -11.466384806913
7.254009628787 -9.371249810768      10.395202849344 -7.215004916358
```